ChatGPT

Rescue Assessment for a Horse-Racing Exchange Betting System

Key findings and diagnosis

The observed realised ROI of **−9.04%** while placing **~1.68M bets** is consistent with a system that is **effectively participating in the market without durable edge** and therefore mostly paying transaction costs. On Australian racing markets, Betfair [1] typically charges commission via a **Market Base Rate (MBR)** that is commonly **~8%** and can be **10%** depending on state and code, with commission taken from **net winnings on a market**. [2] A "bet almost everything" policy tends to convert "no edge" into "negative ROI roughly equal to costs", because you are no longer selecting where you have information advantage, you are simply paying the exchange's fee structure and (usually) the spread/market impact.

The reported **expected ROI of +9.55%** alongside **negative realised ROI in the same sample** is not explainable by commission alone. The gap is large enough that at least one of the following must be true: (a) the EV computation is not aligned to exchange settlement and execution reality, (b) probabilities are catastrophically wrong in the tails (even if average ECE looks fine), or (c) the backtest uses inconsistent prices (e.g., EV uses a stale/optimistic price, realised uses a more pessimistic fill such as BSP or LTP). Bet settlement rules like non-runner reductions and dead-heats can also create systematic drift between naive EV and actual P&L if not modelled at the market level. [3]

The presence of **extreme predictions** (e.g., **$\hat{p}$ = 1.0** for selections trading at **300–890** decimal odds) is a bright-red failure signal. At those odds, the market-implied probability is on the order of **0.1%–0.3%**, so $\hat{p}$ = 1.0 implies either (i) leakage or misalignment (features encode outcome or post-event information), or (ii) a numerical/saturation artefact that your downstream EV code treats as literal certainty and therefore manufactures absurd "value". Even if the "average" calibration metric looks good, rare but extreme probability errors can dominate economic outcomes when the strategy sizes or selects based on EV. The fact that you are currently "betting almost everything" amplifies this, because there is no portfolio-level control preventing tail mistakes from becoming systematic losses.

A salvage path exists, but only if you treat the market price as the baseline forecast and restrict the model to **small, demonstrably stable corrections** and **sparse execution**. That principle mirrors how successful quantitative wagering accounts have historically framed the problem: combine public implied probabilities with an independent handicapping signal, rather than trying to out-predict the market in absolute terms. [4] If that "incremental correction" cannot produce positive edge after realistic costs on strictly out-of-sample walk-forward tests, the correct action is to stop.

Why expected ROI diverges from realised ROI on an exchange

A large expected-versus-realised divergence in exchange betting has a small set of recurring mechanical causes. Your symptoms match several.

Commission is market-level and state-dependent, not a fixed per-bet haircut

Betfair charges commission on **net market winnings**, applying the MBR to the net winnings after the market settles; if you lose on a market, you pay **no commission**. [5] This matters operationally because if you place multiple bets in the same race/market (especially "almost everything"), the realised commission depends on the netted outcome distribution across that market, not the per-bet win/loss. Naively subtracting "8% of each winning bet" is already a mismatch; ignoring the interaction between wide coverage and commission creates systematic EV inflation when the system behaves like a near-dutch.

Settlement adjustments can silently break EV if not simulated exactly

In horse racing exchange markets, non-runners trigger **reduction factors** that alter either traded prices (win markets) or potential winnings (place markets), and dead-heats split stakes/returns according to the number of winners versus the number of "expected winners". [6] If your EV uses the displayed traded price while your realised P&L is affected by reductions, your EV is biased upward. The direction of this bias depends on withdrawals and market type, which means a "simple average" correction is not enough; you need rule-consistent settlement at the bet level and netting at the market level.

Price-time inconsistency is the most common EV bug in market-driven models

Betfair Starting Price is determined at the start of the event from the interaction between SP requests and unmatched exchange orders; it is explicitly a **start-time construct**, not "the best price you could have gotten earlier". [7] If expected ROI is computed using "best available odds at time t" but realised ROI is settled using BSP or a later traded price, you are comparing two different markets. This can easily create "positive expected, negative realised" even with decent calibration.

Execution and fill assumptions can turn a theoretical edge into negative realised P&L

On an exchange, you can see odds and available liquidity, but you may not be fully matched at that price. Betfair explicitly notes that limited liquidity at best odds can prevent full matching, and unmatched bets can lapse if not matched before market state changes. [8] If your backtest assumes full size at the top-of-book "available to back" price, it is optimistic unless you also model queue priority and depth consumption. The developer documentation confirms that exchange market data is an order-book ladder (prices and amounts), reinforcing that execution is a microstructure problem, not a single scalar price. [9]

Tail probability mistakes can dominate EV at long odds while barely moving ECE

Expected Calibration Error is a binned average of absolute calibration gaps. It is widely documented that ECE can behave pathologically under static binning and can hide serious miscalibration via cancellation effects within bins. [10] Reliability diagram stability and calibration assessment choices materially change conclusions; methods like CORP were proposed to avoid ad hoc binning artefacts. [11] Even a small absolute error at long odds (e.g., 0.2% vs 0.1%) can swing EV by large multiples of stake, so economic performance can collapse while global ECE remains low.

Backtest overfitting and multiple testing can produce "beautiful expected value" for fragile patterns

Systematic research tends to select strategies after many trials (features, hyperparameters, time windows, filters). Selection bias and backtest overfitting are well-studied in quantitative finance, including formal tools for the probability of overfitting and the unreliability of repeated "backtest until it looks good". [12] Your setup (multiple market types, many features, huge bet count) is particularly exposed if strategy selection iterated on the same historical period.

Known failure modes in horse-racing ML systems

Horse racing has several “special” failure modes that do not show up as cleanly in IID classification benchmarks.

Market price is already a strong forecast; using it as a dominant feature often yields “cost-only” performance

Betting exchanges and related wagering markets are frequently found to be informationally efficient relative to traditional bookmaker markets, with reduced biases and improved pricing. [13] In efficient prediction markets, the market price tends to be among the best available forecasts, and systematically beating it requires information not already reflected in the price. [14] When feature importance is dominated by market variables and the strategy bets broadly, it is common for realised performance to converge toward “market return minus costs”.

Favourite–longshot bias concentrates pain in the long tail

The favourite–longshot bias is a longstanding empirical regularity in horse wagering where longshots tend to be overbet and favourites underbet, leading to worse average returns at longer odds. [15] Evidence suggests this bias is reduced but not eliminated in exchange markets compared to traditional markets. [16] A model that produces overconfident longshot probabilities – even slightly – will systematically manufacture positive EV that does not exist.

Multi-entry coherence is compulsory; independent per-runner probabilities can be internally inconsistent

Win outcomes in a race are mutually exclusive, which makes the natural object a **race-level distribution** over horses. Classical horse race probability models explicitly treat races as multi-entry contests, including formulations for ordering probabilities and related multi-outcome structures. [17] If you train independent binary models per runner without enforcing race-level constraints, you can generate incoherent probability mass ($\mathrm{sum}(\hat{p}) \neq 1$) and unstable tails. That incoherence has an economic consequence: your portfolio can accidentally “buy” too much probability mass in a race when you bet many runners.

Leakage is easy to introduce through “timestamp ambiguity” in racing data feeds

Betting features that reference “starting price”, “last traded price at off”, “post-scratch adjusted price”, or “final market percentage” can leak the market’s final aggregation of information into earlier decisions. Betfair’s own documentation makes clear that BSP is computed at the start and can be revised with reduction factors under certain conditions; it is not a generic pre-race feature unless your decision time is exactly “at the jump”. [18] Any feature computed with a hidden “after off” dependency creates an illusion of predictive power that evaporates in live execution.

Exchange rule complexity creates subtle P&L drift if ignored

Non-runner reductions differ by market type (win vs each-way vs place), and dead-heats alter stakes/returns via explicit exchange rules. [6] Systems that ignore these mechanics often show “good” model performance but unexpected negative economics after settlement.

Robust alternatives for probability calibration and EV estimation at long odds

The core requirement is not “a lower ECE”. The core requirement is **conservative, tail-safe decision probabilities** whose errors are small relative to the break-even probability implied by odds after costs.

Proper scoring rules and reliable diagnostics should replace “ECE-first” comfort
Proper scoring rules assess probabilistic forecasts in a way that is incentive-compatible for honest probabilities (strict propriety) and are the standard evaluation tool in forecast verification. [19] ECE is widely used as a diagnostic, but it has known pathologies and can be highly sensitive to binning design choices. [10] Stable reliability diagram methods such as CORP were proposed to reduce ad hoc artefacts and provide reproducible calibration assessment. [20]
A practical implication for your case: if you do not measure tail calibration in a way that weights economically relevant regions, you will keep finding “well-calibrated” models that lose money.

Calibration methods worth prioritising in exchange wagering
Tree ensembles and boosted models often need post-hoc calibration when the downstream decision requires probability accuracy. This is a long-established empirical finding (e.g., boosted trees become strong probability predictors after calibration), and modern work reiterates that calibration is often necessary. [21]
The calibrators most relevant to your long-odds failure mode are those that: (a) do not force a single sigmoid shape, (b) behave sensibly at extremes, and (c) can be fitted out-of-fold.

The following are technically grounded candidates:

- **Beta calibration**: designed as a richer, well-founded alternative to logistic calibration; it can include the identity function and tends to behave better when a sigmoid is not the right shape. [22]
- **Bayesian Binning into Quantiles (BBQ)**: model-averaged binning designed to reduce sensitivity to bin choice and improve robustness relative to single histograms. [23]
- **Venn–Abers predictors (including inductive Venn–Abers)**: produce probability-type predictions with guarantees of calibration under exchangeability assumptions, and often provide a range of probabilities that can be used as an uncertainty envelope. [24]
- **Isotonic regression**: useful but can overfit in sparse extreme regions; it benefits from careful data partitioning and, in many cases, ensembles/regularisation (e.g., BBQ, near-isotonic variants). [25]
- **Temperature/Platt-style scaling**: simple and effective in some regimes, but often too restrictive if your miscalibration is asymmetric in tails. [26]

Market-anchored shrinkage is the most important “tail safety” mechanism
A calibration map alone will not stop absurd $\hat{p}$ unless you also constrain the model’s degrees of freedom relative to the market. The most robust pattern in horse-racing wagering research is to treat the market’s implied probability as a strong baseline and model an increment. A historically influential formulation explicitly combines a fundamental model with the public implied probability via a logit-based combination. [27]

A concrete instantiation that is appropriate for exchange betting:

- Define a market-implied probability at decision time t:

$$p_{mkt} = \frac{1}{o_{back}}$$

(or a normalised race-level probability if you have the full book; see below).

- Define your model's raw score s (logit of raw p̂ or the LightGBM margin).
- Fit a **stacking calibration** on out-of-fold predictions:

$$\text{logit}(p) = \alpha + \beta \cdot \text{logit}(p_{mkt}) + \gamma \cdot s$$

with regularisation and explicit caps on γ if needed.

This "market as prior" construction makes it mechanically difficult for the model to claim $p \approx 1$ when the market implies $p \approx 0.001$, unless the data provide overwhelming and repeatable evidence. It also aligns with the economic reality that exchange prices aggregate a large amount of public information. [28]

Odds-banded calibration and EV haircuts are mandatory for longshots
Long-odds regions are data-sparse in terms of positive class outcomes. Even with millions of rows, the count of winners at odds 300+ is small relative to the number of runners trading there; this makes calibrator variance large and creates a strong incentive to overfit tails.

A tail-safe approach uses two layers:

- **Band the calibrator by odds** (e.g., separate mappings for odds buckets such as [1–3], (3–10], (10–30], (30–100], (100+]), and fit each band on out-of-fold data). This forces the model to learn that the mapping from score to probability differs materially across odds, which is consistent with favourite–longshot bias and tail behaviour. [29]
- **Use a conservative probability for betting decisions**, not the point estimate. With Venn–Abers, you can use the lower bound of the probability interval as p_{low} for EV. [30] With other calibrators, you can approximate p_{low} via bootstrap over races/markets or via an ensemble over folds.

Compute EV using exchange-consistent net returns
For a back bet at decimal odds o with stake 1 and commission rate c, the per-bet net profit if it wins is approximately $(o-1)(1-c)$ and if it loses is -1, assuming the bet is the only position in that market. Betfair confirms commission is taken from net winnings via the MBR. [31]
A conservative single-bet EV proxy is:

$$\text{EV}_{net} \approx p_{low}(o-1)(1-c) - (1-p_{low})$$

This is intentionally conservative because real commission is market-level netted and settlement adjustments (reductions, dead heats) can further reduce profits. [32]

If you cannot make EV_{net} positive under conservative assumptions, you do not have a tradable edge.

Best-practice walk-forward backtesting protocol for this setup

A rescue plan lives or dies on the backtest. A walk-forward protocol for exchange wagering must explicitly define **decision time**, **information set**, **execution**, and **market-level settlement**.

Define the decision snapshot and freeze the feature set to it

Pick a small set of decision times relative to the scheduled off (e.g., T−30 min, T−10 min, T−2 min, or "BSP at off"). Treat each decision time as a different strategy because the microstructure and efficiency regime differ across time. Betfair's own educational material highlights that market "percentage" and liquidity change over time, implying that price formation and execution conditions are time-dependent. [33]

If you cannot guarantee that every one of your 131 features is computable from information available at your chosen decision time, the backtest is invalid.

Use race-grouped, strictly time-ordered splits; add embargo if any labels overlap a feature window

Horse racing labels occur at race completion, so "label horizon overlap" is not the same as in finance, but leakage still happens through shared events and ambiguous timestamps. The correct unit for splitting is the **race/market**, not the runner row.

A concrete protocol:

- **Outer walk-forward**: train on $[t_0, t_k]$, validate on $(t_k, t_{k+h}]$, test on $(t_{k+h}, t_{k+2h}]$. Roll forward by h (e.g., 2–4 weeks).
- **Group constraint**: all runners within the same race must stay in the same fold.
- **Embargo**: exclude a small buffer around the fold boundary if you compute features from a moving time window that might straddle races or use meeting-level aggregates (purging/embargo concepts from financial ML are directly motivated by preventing overlap between training information and test labels). [34]

Even if you do not implement full CPCV, the principle of leakage control should be enforced because backtest overfitting and repeated tuning are known to produce false discoveries. [12]

Calibrate only on past data; never calibrate on the test fold

Fit the calibrator on validation predictions generated from models trained without the validation data (true out-of-fold). Calibration methods are explicitly intended as post-processing steps requiring held-out data; many calibration failures come from reusing training data. [35]

Model execution with one of two defensible fill assumptions

Execution dominates your problem because the model is currently "betting almost everything".

Two defensible backtest modes:

- **BSP-only backtest (recommended for first rescue iteration)**
  BSP is a single price determined at the start of the event. It reduces execution uncertainty relative to limit-order simulation and is considered a conventional and simple approach in Betfair's own data-science tooling discussions of backtesting, precisely because it avoids modelling matching uncertainty. [36]
  Use this mode to answer: "Does any signal exist after costs if I assume I can transact at BSP (or a BSP-like price)?"

- **Order-book backtest (required before live deployment)**
  Use the "available to back/lay" ladder and available sizes as supplied by the API/stream to simulate

fills at your intended order type and size. Betfair's developer materials describe that the ladder includes prices and the amounts available at each, and educational material notes limited liquidity can prevent full matching. [37]
Include a probabilistic fill model if you do not have queue position data, but pessimistically assume partial fills at the best price and spillover down the ladder when size exceeds displayed liquidity.

Apply settlement rules and commission exactly
At minimum:

- Apply MBR commission on **net winnings per market**, consistent with Betfair's description. [38]
- Apply non-runner reduction factors exactly as specified for win and place markets. [39]
- Apply dead-heat adjustments using the exchange's described proportional stake reduction. [40]

Include operational costs that "bet almost everything" triggers
If a live implementation places and cancels substantial numbers of orders, Betfair may impose transaction charges after high transaction counts; "transactions" include unmatched, matched, cancelled, and lapsed bets. [41] With a near-universal betting strategy, this is not a theoretical footnote; it can become a real drag.

Selection and risk controls that fit exchange betting realities

Your current strategy is effectively unbounded selection. The rescue requires explicit controls that make the system sparse, tail-safe, and execution-aware.

Selection must be based on conservative edge after costs and uncertainty
A practical decision rule for a back bet:

1. Compute p_{low} (Venn–Abers lower probability or an ensemble-based lower bound). [30]
2. Compute EV_{net} using commission c and a conservative fill price o. [42]
3. Bet only if:
4. $\mathrm{EV}_{net} \geq \tau$ where τ is a positive margin (not 0), and
5. the available liquidity at o exceeds your stake by a safety multiple (e.g., $\geq 5\times$ stake), reflecting partial fill risk. [43]

Given Australian racing commission is often ~8–10%, a reasonable starting τ is **+1% to +3%** per bet after costs, because a razor-thin margin will be consumed by microstructure and modelling error even if you are "right on average". [44]

Odds-banding is a risk control, not a "betting preference"
Until tail calibration is proven, impose explicit odds caps and staged expansion:

- Phase A: allow odds **≤ 30** only.
- Phase B: allow odds **≤ 100** only if the odds-band calibration passes reliability checks and out-of-sample profit tests in Phase A.
- Phase C: allow odds **> 100** only if you can demonstrate stable tail calibration and positive net EV under conservative assumptions.

This is aligned with the empirical reality that longshot regions behave differently and are affected by favourite–longshot bias. [29]

Per-race exposure limits are compulsory in multi-entry markets
If you place multiple bets in a single race, you create correlated exposures and interact with commission netting. Define a maximum total stake per race and per market type (win/place/etc.). Settlement mechanics like dead-heats and reduction factors also apply at the market level, reinforcing that the market is the natural unit. [45]

Bet sizing should be fractional Kelly with hard caps
Kelly sizing remains the canonical result for maximising long-run growth under stated assumptions, but it is notoriously brittle under probability error. John L. Kelly Jr. [46] derived the criterion in the context of favourable betting odds. [47] In model-driven wagering, use **fractional Kelly** (e.g., 5%–20% Kelly) and add a cap per bet and per race. This is not optional when your model is still showing tail pathologies.

A practical sizing rule:

- Compute f_{kelly} from p_{low}, o, and cost-adjusted payoff.
- Place stake $= \min(f_{kelly} \cdot \lambda, f_{max}) \cdot bankroll$, with $\lambda \in [0.05, 0.2]$ and f_{max} an absolute cap (e.g., 0.25% bankroll per bet, 1% per race) until proven.

Continue vs stop decision rule, ranked interventions, and an experiment plan

Continuation is justified only if the system can demonstrate **repeatable incremental edge over the market after costs**, under walk-forward evaluation, without tail blow-ups. Otherwise, stop.

Top ten interventions ranked by impact, effort, and confidence

Intervention	Why it targets your failure mode	Impact	Effort	Confidence
Rebuild the backtest to match exchange settlement (commission netting, reductions, dead-heats)	Eliminates systematic EV inflation from rule mismatch	Very high	Medium	High [48]
Enforce time-true snapshots and price alignment (decision-time odds = EV odds = fill odds)	Removes the most common "expected vs realised" inconsistency	Very high	Medium	High [49]
Stop "bet almost everything": introduce conservative edge thresholds and odds bands	Prevents cost-only participation and tail-driven overbetting	Very high	Low	High [50]

Intervention	Why it targets your failure mode	Impact	Effort	Confidence
Market-anchored shrinkage (logit(p) anchored to logit(p_mkt))	Directly prevents absurd $\hat{p}$ at long odds; aligns with known successful practice	Very high	Medium	High [51]
Replace ECE-only monitoring with tail-weighted calibration diagnostics and proper scoring rules	ECE can hide tail failures; proper scoring rules provide principled evaluation	High	Medium	High [52]
Use Venn–Abers or BBQ/ beta calibrators with odds-banded fitting	Reduces tail overconfidence and provides uncertainty bounds	High	Medium	Medium–high [53]
Switch model target to "residual vs market" instead of absolute probability	Prevents the model from trying to outpredict the market baseline directly	High	High	Medium [54]
Implement order-book execution simulation or BSP-only gating as a staged deployment	Prevents paper alpha from vanishing due to fill assumptions	High	Medium–high	Medium–high [55]
Add bankroll rules: fractional Kelly, per-race caps, kill-switch drawdowns	Converts model uncertainty into bounded risk	Medium	Low–medium	High [47]
Add overfitting controls: strict walk-forward, limited tuning, overfit diagnostics (CSCV/PBO)	Reduces the probability you are chasing noise	Medium–high	High	Medium [56]

Four-week experiment plan with hypotheses, metrics, and pass/fail thresholds

Week 1: Integrity and economics reconciliation (no new modelling yet)
Hypothesis: The +9.55% expected ROI is largely an artefact of EV/backtest mismatch, not genuine edge.
Work:
- Implement market-level settlement: commission on net winnings; non-runner reductions; dead-heat logic. [48]
- Enforce decision-time price consistency and verify that every feature is available at the decision snapshot. [57]

Metrics and thresholds:
- **Pass** if absolute gap $|\text{expected ROI} - \text{realised ROI}|$ on the same bets shrinks to ≤ **2.0%** after rule-consistent EV, or is explained by explicitly measured execution slippage.
- **Fail** if the gap remains **> 2.0%** and is not attributable to execution, indicating probability estimates or joins are fundamentally broken.

Week 2: Tail-safe probability layer (calibration + market anchoring)
Hypothesis: Market-anchored shrinkage plus odds-banded calibration eliminates absurd tail EV and reduces overbetting pressure without killing all signal.
Work:
- Build a stacking calibrator anchored to market implied probability. [54]
- Fit odds-banded calibrators (beta calibration or BBQ; optionally Venn–Abers for uncertainty bounds). [58]
Metrics and thresholds:
- **Pass** if no calibrated probabilities exceed a hard sanity constraint such as $p > 0.10$ when odds > 50, and p never equals exactly 0 or 1 after clipping. (The specific cap can be data-driven later; this is a safety gate.)
- **Pass** if tail diagnostic shows monotone calibration behaviour in the long-odds bands (via stable reliability assessment rather than a single ECE). [59]
- **Fail** if extreme p̂ behaviour persists in long-odds bands or if calibration collapses (e.g., produces worse log loss/Brier than the market baseline). [60]

Week 3: Strategy sparsification and realistic execution
Hypothesis: After costs and conservative probabilities, a sparse strategy can exceed cost-only performance; if it cannot, there is no tradable edge.
Work:
- Introduce edge thresholds τ and odds bands; cap per-race exposure; fractional Kelly sizing. [61]
- Evaluate in BSP mode first; then order-book simulation for the final candidate. [62]
Metrics and thresholds (walk-forward out-of-sample only):
- **Pass** if net ROI (after commission and settlement) is **> +1.0%** with a confidence interval that includes a meaningful buffer over zero when computed at the market/race level (clustered).
- **Pass** if the number of bets falls by at least **90%** versus the current “bet almost everything” regime while retaining non-trivial turnover.
- **Fail** if ROI remains ≤ 0 under conservative fills and costs, or if performance depends entirely on longshot exposure.

Week 4: Robustness and stop/continue decision
Hypothesis: Any real edge should persist across multiple walk-forward folds and not be a single-regime artefact.
Work:
- Run the final candidate through multiple contiguous walk-forward segments, with no retuning between segments.
- If you have tried many variants, compute an explicit overfitting diagnostic such as CSCV/PBO on the candidate pool before declaring success. [63]
Metrics and thresholds:
- **Pass** if ROI is positive in at least **70%** of fold segments and the worst-segment ROI is not catastrophically negative (e.g., worse than −5% after costs), indicating stability rather than luck.
- **Fail** if fold-to-fold variance is extreme or PBO is high (e.g., **PBO ≥ 0.5**), suggesting the “winner” is likely an artefact of selection. [63]

Explicit kill criteria for shutting down the horse-racing effort

Kill criteria should be objective and tied to the specific failure modes you are observing:

- **Integrity kill**: Any verified leakage or timestamp ambiguity where features depend on information available only at/after the off (e.g., BSP-derived fields in a T−10 strategy) results in an immediate stop until the data contract is fixed. BSP is explicitly defined at the start of the event, so it cannot be a pre-off feature without redefining decision time. [7]
- **Tail safety kill**: If, after calibration and market anchoring, the system still emits probabilities that imply near-certainty on high-odds runners (e.g., $p > 5\%$ at odds > 100) in out-of-sample data, stop. This is a risk control, not a "tuning choice".
- **Economics kill**: After Week 3, if the best conservative strategy (BSP mode, after rule-consistent settlement and commission) cannot achieve **positive net ROI** in out-of-sample walk-forward testing, stop. The market-efficiency literature and the dominance of market features in your models imply that without demonstrable incremental edge, you are paying costs for no information advantage. [64]
- **Overfitting kill**: If the "winning" configuration emerges only after extensive specification search and overfitting diagnostics suggest high risk (e.g., CSCV-based PBO $\geq$ 0.5), stop or restart with a stricter research protocol and a fresh holdout period. [56]
- **Operational kill**: If the live implementation would require order volumes that plausibly trigger transaction charges or chronic partial fills (because liquidity at the price is consistently below stake), stop. Transactions include unmatched/cancelled/lapsed bets; high-transaction strategies have explicit cost/constraint implications on Betfair. [65]

The stop recommendation becomes final if you fail the Economics kill **after** fixing the integrity and tail-safety layers, because at that point the most plausible explanations are (i) the market signal already prices the information you have, and (ii) remaining model "edge" is noise that will not survive costs.

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[1] [30] [53] https://proceedings.neurips.cc/paper/2015/file/a9a1d5317a33ae8cef33961c34144f84-Paper.pdf
https://proceedings.neurips.cc/paper/2015/file/a9a1d5317a33ae8cef33961c34144f84-Paper.pdf

[2] [44] [50] [61] https://www.betfair.com.au/hub/education/understanding-commission/
https://www.betfair.com.au/hub/education/understanding-commission/

[3] [6] [32] [39] [45] https://support.betfair.com/app/answers/detail/exchange-horse-racing-rules/
https://support.betfair.com/app/answers/detail/exchange-horse-racing-rules/

[4] [27] [51] [54] https://gwern.net/doc/statistics/decision/1994-benter.pdf
https://gwern.net/doc/statistics/decision/1994-benter.pdf

[5] [31] [38] https://www.betfair.com.au/aboutUs/Betfair.Charges/
https://www.betfair.com.au/aboutUs/Betfair.Charges/

[7] [18] [49] [57] https://www.betfair.com.au/hub/education/betfair-basics/betfair-starting-price-bsp/
https://www.betfair.com.au/hub/education/betfair-basics/betfair-starting-price-bsp/

[8] [33] [43] https://www.betfair.com.au/hub/education/betfair-basics/read-an-exchange-market/
https://www.betfair.com.au/hub/education/betfair-basics/read-an-exchange-market/

9 37 https://support.developer.betfair.com/hc/en-us/articles/6540502258077-What-Betfair-Exchange-market-data-is-available-from-listMarketBook-Stream-API
https://support.developer.betfair.com/hc/en-us/articles/6540502258077-What-Betfair-Exchange-market-data-is-available-from-listMarketBook-Stream-API

10 52 https://openaccess.thecvf.com/content_CVPRW_2019/papers/Uncertainty%20and%20Robustness%20in%20Deep%20Visual%20Learning/Nixon_Measuring_Calibration_in_Deep_Learning_CVPRW_2019_paper.pdf
https://openaccess.thecvf.com/content_CVPRW_2019/papers/Uncertainty%20and%20Robustness%20in%20Deep%20Visual%20Learning/Nixon_Measuring_Calibration_in_Deep_Learning_CVPRW_2019_paper.pdf

11 20 59 https://www.pnas.org/doi/10.1073/pnas.2016191118
https://www.pnas.org/doi/10.1073/pnas.2016191118

12 https://www.ams.org/notices/201405/rnoti-p458.pdf
https://www.ams.org/notices/201405/rnoti-p458.pdf

13 16 64 https://irep.ntu.ac.uk/id/eprint/10053/1/196111_155%20Vaughan%20Williams%20Post%20Print.pdf
https://irep.ntu.ac.uk/id/eprint/10053/1/196111_155%20Vaughan%20Williams%20Post%20Print.pdf

14 28 https://jmvidal.cse.sc.edu/library/wolfers04a.pdf
https://jmvidal.cse.sc.edu/library/wolfers04a.pdf

15 29 https://www.nber.org/system/files/working_papers/w15923/w15923.pdf
https://www.nber.org/system/files/working_papers/w15923/w15923.pdf

17 https://www.jstor.org/stable/2284068
https://www.jstor.org/stable/2284068

19 60 https://sites.stat.washington.edu/raftery/Research/PDF/Gneiting2007jasa.pdf
https://sites.stat.washington.edu/raftery/Research/PDF/Gneiting2007jasa.pdf

21 35 https://www.cs.cornell.edu/~alexn/papers/calibration.icml05.crc.rev3.pdf
https://www.cs.cornell.edu/~alexn/papers/calibration.icml05.crc.rev3.pdf

22 58 https://proceedings.mlr.press/v54/kull17a/kull17a.pdf
https://proceedings.mlr.press/v54/kull17a/kull17a.pdf

23 25 https://www.dbmi.pitt.edu/wp-content/uploads/2022/10/Obtaining-well-calibrated-probabilities-using-Bayesian-binning.pdf
https://www.dbmi.pitt.edu/wp-content/uploads/2022/10/Obtaining-well-calibrated-probabilities-using-Bayesian-binning.pdf

24 https://arxiv.org/pdf/1211.0025
https://arxiv.org/pdf/1211.0025

26 https://arxiv.org/pdf/1706.04599
https://arxiv.org/pdf/1706.04599

34 46 https://www.garp.org/hubfs/Whitepapers/a1Z1W0000054x6lUAA.pdf
https://www.garp.org/hubfs/Whitepapers/a1Z1W0000054x6lUAA.pdf

36 55 62 https://betfair-datascientists.github.io/tutorials/backtestingRatingsTutorial/
https://betfair-datascientists.github.io/tutorials/backtestingRatingsTutorial/

[40] https://www.betfair.com.au/en/aboutUs/Rules.and.Regulations/
https://www.betfair.com.au/en/aboutUs/Rules.and.Regulations/

[41] [65] https://support.developer.betfair.com/hc/en-us/articles/20029343399836-Transaction-Charge-how-are-transactions-counted
https://support.developer.betfair.com/hc/en-us/articles/20029343399836-Transaction-Charge-how-are-transactions-counted

[42] [48] https://www.betfair.com.au/hub/faq/
https://www.betfair.com.au/hub/faq/

[47] https://www.princeton.edu/~wbialek/rome/refs/kelly_56.pdf
https://www.princeton.edu/~wbialek/rome/refs/kelly_56.pdf

[56] [63] https://www.davidhbailey.com/dhbpapers/backtest-prob.pdf
https://www.davidhbailey.com/dhbpapers/backtest-prob.pdf